Foundations of Programming

Assignment No.: 18

PROBLEM STATEMENT: Write a C program that accepts a string from the user and performs the following string operations: i) Calculate length of string ii) String reversal iii) Equality check of two strings iv) Check palindrome v) Check substring.

OBJECTIVE:

- To understand basic string operations in C.
- To learn how to manipulate strings using loops and functions.
- To perform comparison, reversal, and substring checking.
- To check whether a string is a palindrome.

THEORY:

In C programming, a string is a sequence of characters terminated by a null character ('\0'). Strings are commonly stored in character arrays. Various operations can be performed on strings such as calculating their length, reversing the characters, comparing two strings, checking whether a string is a palindrome, and finding a substring within another string. These operations can be implemented using loops and conditional statements or by using standard library functions from the string.h header file.

PLATFORM: Linux

INPUT: A string and another string for comparison and substring checking.

```c
#include <stdio.h>
#include <string.h>

// Function to reverse string
void reverseString(char str[]) {
    int i, len = strlen(str);
    char rev[100];

    for(i = 0; i < len; i++) {
        rev[i] = str[len - i - 1];
    }
    rev[len] = '\0';

    printf("Reversed String: %s\n", rev);
}

// Function to check palindrome
void checkPalindrome(char str[]) {
    int i, len = strlen(str), flag = 1;

    for(i = 0; i < len/2; i++) {
        if(str[i] != str[len - i - 1]) {
            flag = 0;
            break;
        }
    }

    if(flag)
        printf("It is a Palindrome\n");
    else
        printf("Not a Palindrome\n");
}

// Function to check substring
void checkSubstring(char str[], char sub[]) {
    if(strstr(str, sub))
        printf("Substring found\n");
    else
        printf("Substring not found\n");
}

int main() {
    char str1[100], str2[100], sub[100];

    // Accept main string
    printf("Enter a string: ");
    gets(str1);

    // i) Length
    printf("Length of string: %lu\n", strlen(str1));

    // ii) Reverse
    reverseString(str1);

    // iii) Equality check
    printf("Enter another string for comparison: ");
    gets(str2);

    if(strcmp(str1, str2) == 0)
        printf("Strings are equal\n");
    else
        printf("Strings are not equal\n");

    // iv) Palindrome
    checkPalindrome(str1);

    // v) Substring
    printf("Enter substring to search: ");
    gets(sub);

    checkSubstring(str1, sub);

    return 0;
}
```

OUTPUT: Displays string length, reversed string, equality result, palindrome result, and substring check result.

```
Enter a string: madam
Length of string: 5
Reversed String: madam
Enter another string for comparison: mad
Strings are not equal
It is a Palindrome
```

SAMPLE INPUT:

Enter a string: madam
Enter another string for equality check: mad

SAMPLE OUTPUT:

Length of string = 5
Reversed string = madam
The string is a Palindrome
Strings are not equal
Substring found in main string

CONCLUSION: Thus, the C program successfully performs different string operations such as length calculation, reversal, equality check, palindrome check, and substring detection.

FAQs:

1. What is a string in C programming?
2. Which header file is required for string functions?
3. What function is used to find the length of a string?
4. What is a palindrome string?
5. Which function is used to check a substring in C?